SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: **DESIGN AND ANALYSIS OF ALGORITHMS**

COURSE CODE: **CSA06**

COURSE OUTCOME COVERED

CO1: Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method.
(BL-1, BL-2, BL-3)

Assessment Tool Weightage: 35%

QUESTIONS: 10 **Total Marks: 50**

Annexure A

Concept Mapping

Code of Conduct:

I ____________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Concept Mapping

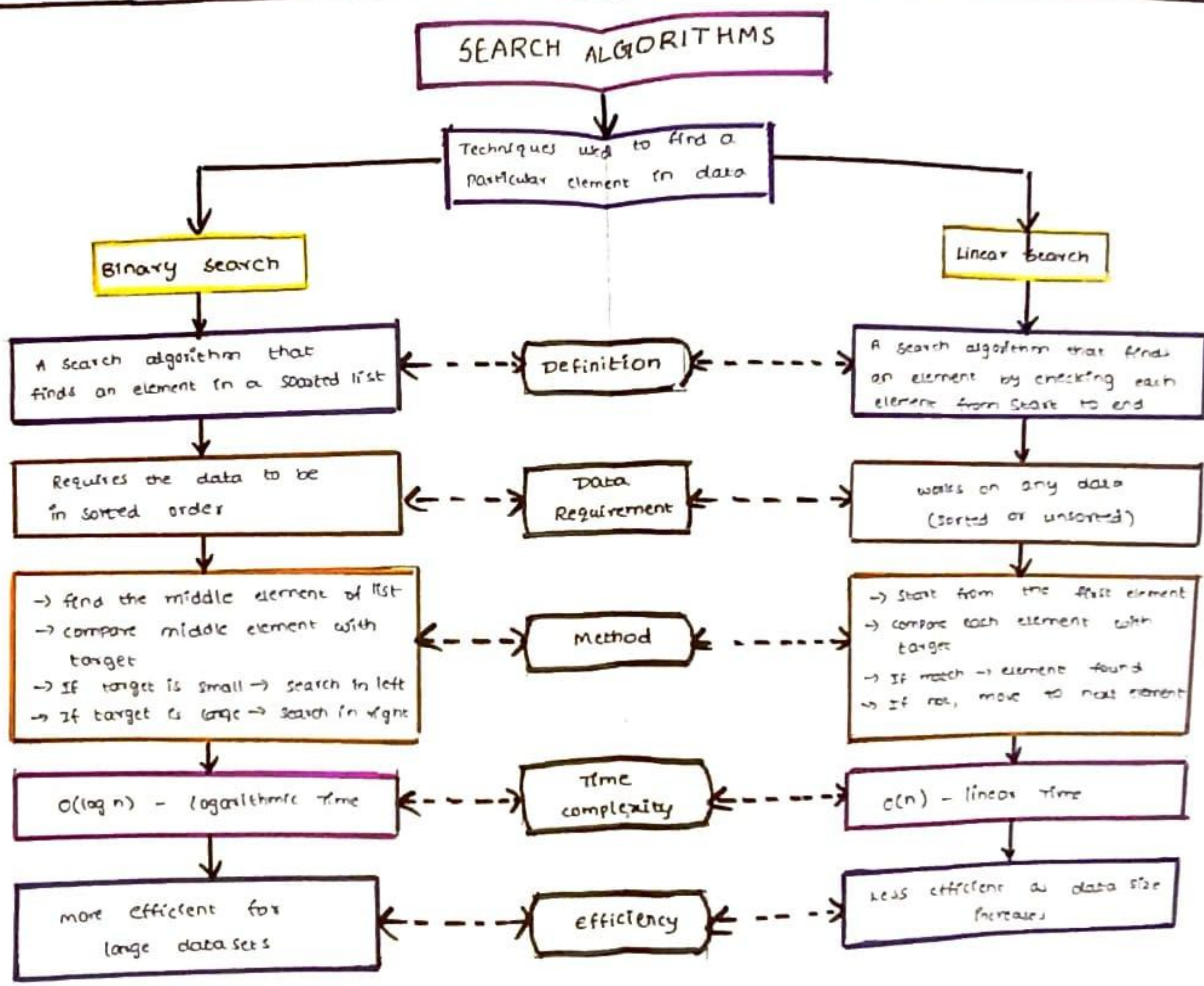
Q6. Consider the concept map: **Search Algorithm → Binary Search → Input Division → Logarithmic Complexity → Efficiency** Identify how each node connects causally to the next. Extend the map with a parallel branch for **Linear Search → O(n)** and explain, through the comparative links, why Binary Search's structure leads to a superior growth rate.

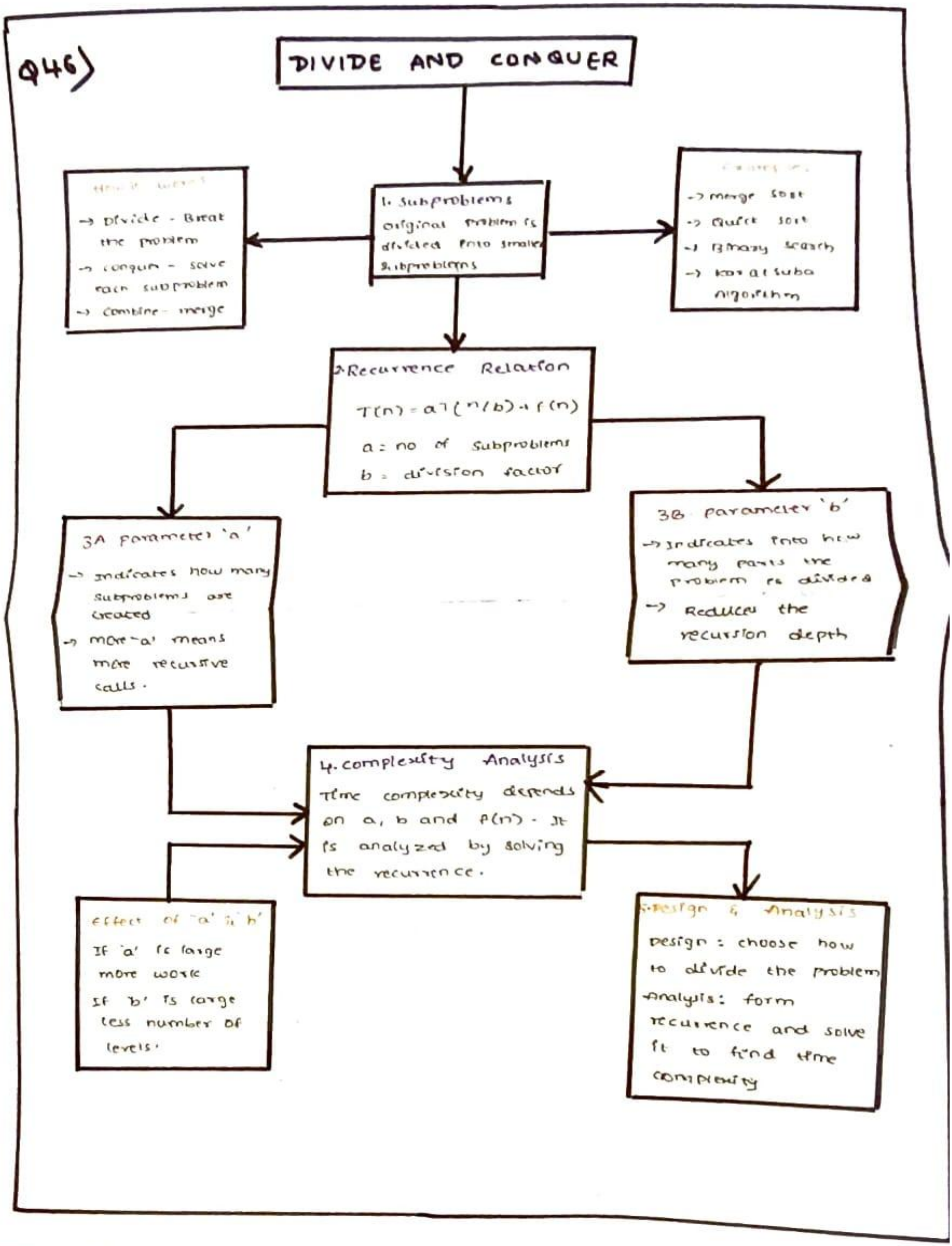
Q46. Consider the concept map: Divide and Conquer → Subproblems → Recurrence → Complexity Map how problem division leads to recurrence. Extend with parameters a and b. Justify their role in complexity. Interpret how this connects design and analysis.

RUBRICS FOR CALCULATION

Criteria	Weight age	Level 4 –	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Concept Identification	25%		Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%		Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%		Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%		Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%		Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

Q) 6


SEARCH ALGORITHMS
Techniques used to find a particular element in data
Binary search
Linear search
A search algorithm that finds an element in a sorted list
Definition
A search algorithm that finds an element by checking each element from start to end
Requires the data to be in sorted order
Data Requirement
works on any data (sorted or unsorted)
→ find the middle element of list
→ compare middle element with target
→ If target is small → search in left
→ If target is large → search in right
Method
→ Start from the first element
→ compare each element with target
→ If match → element found
→ If not, move to next element
O(log n) - logarithmic time
Time complexity
O(n) - linear time
more efficient for large data sets
efficiency
Less efficient as data size increases


Q46)
DIVIDE AND CONQUER
1. Subproblems
original problem is divided into smaller subproblems
→ Divide - Break the problem
→ Conquer - solve each subproblem
→ Combine - merge
→ merge sort
→ Quick sort
→ Binary search
→ Karatsuba algorithm
→ Recurrence Relation
T(n) = aT(n/b) + f(n)
a: no of subproblems
b = division factor
3A Parameter 'a'
→ indicates how many subproblems are created
→ more 'a' means more recursive calls.
3B Parameter 'b'
→ indicates into how many parts the problem is divided
→ Reduces the recursion depth
4. Complexity Analysis
Time complexity depends on a, b and f(n). It is analyzed by solving the recurrence.
Effect of 'a' & 'b'
If 'a' is large more work
If 'b' is large less number of levels.
5. Design & Analysis
Design: choose how to divide the problem
Analysis: form recurrence and solve it to find time complexity